

The cost of a misdirected cue scales with load: invalid-cue attentional reallocation is recoverable at set size four but abolished at set size nine in a recurrent vision-transformer model of covert attention

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Abstract

Covert spatial cues speed and sharpen detection at their target, but a cue that points to the wrong location imposes a cost. We ask how that invalid-cue cost scales with stimulus load in a trained recurrent attention model, and whether it is a property of the top-down read-out specifically. A single convolutional, self-distilling, cross-attention recurrent agent was trained from raw pixels on a value-directed, cued orientation-change-detection task, and probed at two stimulus set sizes (four and nine locations) by reading the attention its memory pathway places on the changed location. At set size four, an invalid cue delays the attentional reallocation to the uncued change by one frame but the spotlight recovers fully thereafter (excess change-lock +0.70 over the uniform baseline), and behaviour confirms that uncued changes are still detected, only at higher magnitude. At set size nine the reallocation is abolished: attention never reaches the uncued change (≈ 0 excess at both frames), even though the cued read-out and valid-cue locking remain intact. A bottom-up element-wise-affine variant is immune, locking on uncued changes at high load but expressing a weaker cueing benefit. The set-size scaling thus traces specifically to the cue-gated top-down channel, yielding a falsifiable behavioural prediction (stated here as a prediction pending convergence) that the invalid-cue cost should grow with set size.

1 Introduction

A spatial cue that reliably indicates where a relevant event will occur speeds and sharpens its detection, while a cue that misdirects attention imposes a measurable cost: detection at the uncued location is slower and less sensitive than it would be with no cue at all (Posner et al., 1980, 1994). This valid-benefit/invalid-cost signature is the canonical operational definition of covert spatial attention and a load-bearing tool across its study (Carrasco, 2011). The cost arises because attention, once committed to the cued location, must be reallocated when the relevant event appears elsewhere, and that reallocation is neither instantaneous nor guaranteed.

How large the invalid-cue cost should be is, on mechanistic grounds, a function of stimulus load. Biased-competition and normalization accounts hold that stimuli within a receptive field or suppressive pool compete, and that attention resolves the competition by biasing the gain of the attended item against its neighbours (Desimone et al., 1995; Reynolds et al., 1999, 2000; Reynolds and Heeger, 2009). Under these accounts, adding competitors deepens the suppressive

field around the cued item, so that reallocating the top-down bias *away* from a strongly favoured cued location toward an uncued event becomes progressively harder as set size grows. A priority-map view, in which selection reflects a commitment by collicular and cortical priority circuitry to the cued location, predicts the same direction: a more strongly committed map is more costly to override (Krauzlis et al., 2018, 2013; Zénon and Krauzlis, 2012). Whether this load-scaling of the invalid-cue cost emerges in a system that learns the task end-to-end, rather than being built in, is open.

We address this in a trained recurrent attention agent that performs a value-directed, spatially cued orientation-change-detection task from raw pixels. A coloured cue marks one location and, by its colour, a reward value (Maunsell, 2004; Luo and Maunsell, 2019; Herman et al., 2020); a change occurs on half of trials, and the agent declares wait or change each frame. The agent routes detection through a memory pathway whose read-out we can interrogate directly, asking on which location it places its attention as the change unfolds. We compare two stimulus set sizes (four versus nine locations) within the same model, and contrast a cue-gated top-down read-out against a bottom-up variant of the same architecture. The result, previewed here, is that the invalid-cue cost scales catastrophically with set size in the top-down read-out: a recoverable one-frame delay at the low set size becomes a categorical failure to attend the uncued change at the high set size, while the bottom-up read-out is immune. This identifies the load-scaling of the invalid-cue cost as a specific property of cue-gated top-down selection (Sprague et al., 2018), and yields a concrete behavioural prediction.

2 Task, models, and set-size manipulation

We study a recurrent, pixel-based agent performing a spatially-cued orientation-change detection task and ask how the cost of an *invalid* cue scales with the number of competing locations. The task is value-directed covert attention (Luo and Maunsell, 2019; Carrasco, 2011): on each trial a coloured cue marks one location and, by its colour, signals an associated reward magnitude (the cue colour-values used were [1, 3, 5]). The cue’s spatial reliability is set by a validity proportion. Each episode unfolds over seven logical frames; on half of trials an oriented stimulus changes its orientation, and when a change occurs it appears at frame 5 (f5), so the change is detectable from f5 onward, with f6 being one frame later. The agent emits a **wait/change** decision on every frame. A **valid** trial is one in which the change occurs at the cued location; an **invalid** trial is one in which the change occurs at an uncued location, with the cue therefore pointing away from the event (Posner et al., 1980, 1994).

The agent is a convolutional JEPA architecture trained end-to-end from raw pixels. A squeeze-and-excitation residual convolutional front-end (no variational encoder) extracts per-location features; a V-JEPA prototype-softmax self-distillation objective provides an auxiliary representational signal; a cross-attention memory module (the cue-gated, top-down “crossattn1” reader) feeds a spatial xLSTM recurrent memory (memory width $d_{\text{mem}} = 128$); and a distributional actor-critic head produces the decision. The cross-attention reader is the priority channel of interest: its queries read out over the spatial memory keys, so the distribution of attention over those keys reports which location the agent is monitoring on each frame.

We manipulate stimulus load by changing only the set size, holding the architecture, auxiliary objective, training regime, and decision head fixed. In the **set size 4** condition (SS4, task vda4) the four stimuli occupy a 2×2 grid (25×25 -pixel patches in a 50×50 image). In the **set size 9** condition (SS9, task vda9) nine stimuli occupy a 3×3 grid (75×75 image), with the cue placed at the top-left location (P0) or the bottom-right location (P8). Because the same model and objective are used at both set sizes, any difference in how the invalid cue is handled is attributable to load rather than to a change in mechanism. As a mechanistic control we additionally examine an element-wise-affine variant of the same conv-JEPA at SS9, in which the memory reader is a bottom-up self-attention rather than the cue-gated cross-attention; this

Invalid-cue attentional reallocation: recoverable at set size 4, abolished at set size 9

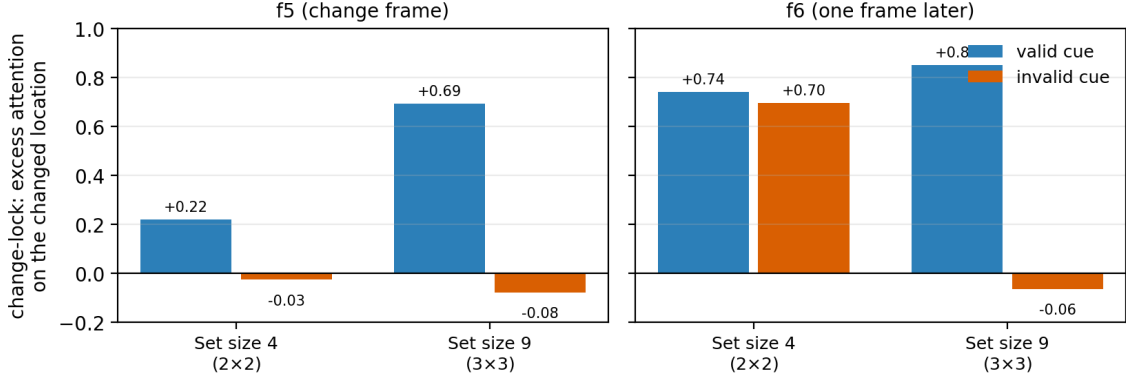


Figure 1. Excess change-lock (attention on the changed location, over the uniform baseline) at frames f5 and f6, for valid and invalid cues, at SS4 (uniform 0.25) and SS9 (uniform 0.111). At SS4 the invalid lock is absent at f5 but recovers fully by f6; at SS9 the invalid lock is abolished at both frames while the valid lock strengthens.

isolates the contribution of the top-down channel.

3 Results

3.1 Set size turns a recoverable invalid-cue delay into a categorical attentional failure

The central result concerns where the cross-attention reader places its attention on the changed location, measured as *excess change-lock*: the fraction of attention on the changed memory key in excess of the uniform baseline (a uniform reader places $1/4 = 0.25$ at SS4 and $1/9 = 0.111$ at SS9). Reporting excess-over-uniform makes the two set sizes directly comparable. Figure 1 summarises the comparison and Table 1 gives the values.

At SS4 (Figure 2), a *valid* cue produces a strong change-lock that is already present at f5 (+0.20 for a cue at S1, +0.24 for a cue at S4) and saturates by f6 (+0.75 / +0.74). On *invalid* trials the lock is *delayed by one frame* rather than lost: at f5 there is no excess attention on the uncued change (−0.03 / −0.02), but by f6 the spotlight has re-pointed to the uncued location and the lock recovers essentially in full (+0.70 / +0.69). At SS4, then, an invalid cue costs the agent a single frame of attentional reallocation.

At SS9 (Figure 3) the valid-trial behaviour is, if anything, stronger: the cross-attention reader locks onto the cued change with excess +0.69 at f5 and +0.86 at f6. The invalid case, however, collapses. Attention never reaches the uncued changed location at either frame: the excess is −0.08 at f5 and −0.06 at f6, i.e. at or slightly below the uniform baseline. The one-frame-late reallocation seen at SS4 does not occur; the model commits to the cued location and is effectively blind to the uncued change. The cue-frame read itself is intact at both set sizes—non-cued queries place essentially all of their attention (≈ 1.00) on the cued image patch, broadcasting the cue location—so the failure is specifically a failure of top-down *reallocation* away from the cued item, not a failure to register the cue. Thus increasing load from four to nine competitors converts the invalid-cue cost from a recoverable one-frame delay into a categorical attentional failure.

Table 1. Excess change-lock (attention on the changed location minus the uniform baseline) on the cross-attention memory keys. SS4 uniform = 0.25; SS9 uniform = 0.111. SS4 values are shown for the two cue placements (S1 / S4); SS9 for the two cue placements (P0 / P8).

Set size	Cue	f5	f6
SS4	valid	+0.20 / +0.24	+0.75 / +0.74
SS4	invalid	−0.03 / −0.02	+0.70 / +0.69
SS9	valid	+0.69	+0.86
SS9	invalid	−0.08	−0.06

RAW cross-attn → MEMORY keys (mean/96); rows=query patch, cols=memory token [S1,TR,BL,S4]; green=cued red=changed — iter 9599

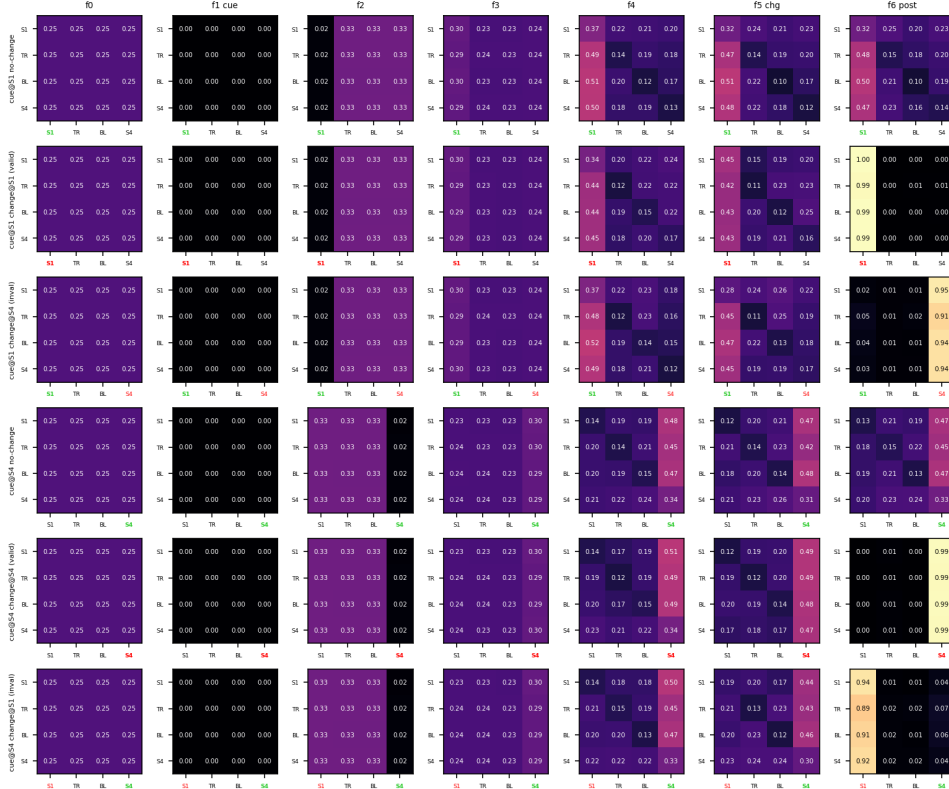


Figure 2. SS4 cross-attention memory maps. Valid cues lock onto the change by f6; invalid cues lock one frame later, with the spotlight re-pointing to the uncued change.

3.2 Behaviour at SS4: an invalid cue shifts the psychometric threshold but does not abolish detection

At SS4 the agent performs the task well. Over 600 greedy episodes it achieves overall correct 0.865, hit rate 0.762, correct-rejection rate 0.956, and false-alarm rate 0.044, with a mean reaction time of 5.11 frames; performance is graded by cue colour-value, with correct rates [0.83, 0.90, 0.86] for values [1, 3, 5]. The valid-versus-invalid psychometric (Figure 4) shows the behavioural signature of the one-frame attentional delay. Across change magnitudes [0, 4, 8, 16, 28, 44, 64] degrees, the valid detection curve is $P(\text{detect}) = 0.05, 0.109, 0.166, 0.584, 0.988, 1.00, 1.00$ and the invalid curve is 0.059, 0.063, 0.100, 0.425, 0.941, 1.00, 0.966. The invalid threshold is shifted rightward (detection is harder): at 16° the valid hit rate of 0.584 exceeds the invalid 0.425, a +0.16 cueing benefit. Crucially, both curves recover to ≈ 1.0 by 44°, so at SS4 uncued changes *are* eventually detected, merely at a higher magnitude—consistent with the attention measurement, in which the invalid lock is delayed but not lost. Reaction times remain modest (~ 5.0 – 5.3 frames) across conditions.

The abolished invalid change-lock at SS9 (Figure 3) generates a concrete behavioural **prediction**: at high load the rightward threshold shift should not merely steepen but should fail to recover, so that uncued changes go largely undetected even at large magnitudes, producing a much larger and less recoverable invalid-cue cost than at SS4. We state this as a prediction to be confirmed at convergence; the SS9 behavioural psychometric is not yet measured.

3.3 A bottom-up reader is immune to the collapse

The catastrophic scaling of the invalid-cue cost is specific to the top-down, cue-gated cross-attention reader. The element-wise-affine variant of the same conv-JEPA at SS9, whose memory reader is bottom-up self-attention rather than cue-gated cross-attention, does *not* show the collapse (Figure 5). On valid trials it locks (+0.71 at f5, +0.66 at f6), and on invalid trials it *still* locks onto the uncued change (+0.61 at f5, +0.59 at f6). Because this reader is driven by stimulus salience, any sufficiently large orientation change pulls attention regardless of where the cue points, so the uncued change is caught even at SS9. The same property makes its cueing benefit weaker—it is not strongly gated by the cue. The contrast localises the load-dependent failure to the top-down channel: a bottom-up salience reader is immune to the invalid-cue collapse but forfeits the selectivity that the cue confers.

3.4 Interpretation

Together these results describe a covert-attention agent that routes detection through a cue-gated top-down channel and therefore pays an invalid-cue cost that scales with load. At low set size the spotlight re-points to the uncued change one frame late, a recoverable delay; at high set size the agent commits to the cued location and the uncued change is missed entirely. This pattern fits biased-competition and normalization accounts of attention (Desimone et al., 1995; Reynolds et al., 1999, 2000; Reynolds and Heeger, 2009; Maunsell, 2004): more competitors in the suppressive field make top-down reallocation away from the cued item progressively harder, until it fails. It also extends the classical Posner finding that the cost of a misdirected cue grows with the number of competing locations (Posner et al., 1980, 1994; Carrasco, 2011), and is consistent with priority-map and commitment accounts in which a selected location is actively maintained (Krauzlis et al., 2013, 2018; Zénon and Krauzlis, 2012; Herman et al., 2020; Sprague et al., 2018). The bottom-up control shows the trade-off cleanly: a salience-driven reader avoids the invalid-cue cost but forfeits the cueing benefit (Luo and Maunsell, 2019).

4 Discussion

The same trained agent expresses two qualitatively different responses to an invalid cue depending on stimulus load. At set size four, an invalid cue costs the model one frame of attentional reallocation: the spotlight, initially committed to the cued location, re-points to the uncued change one frame late and then locks on it fully (excess change-lock +0.70 over the uniform baseline, against +0.75 for a valid cue), and the behavioural psychometric confirms that uncued changes are detected, only at higher orientation magnitude (a rightward threshold shift, e.g. 0.425 versus 0.584 detection at 16°, with both conditions recovering to ceiling by 44°). At set size nine the reallocation does not occur at all: the model places no excess attention on the uncued change at either frame (≈ 0 / slightly negative), even though valid-cue locking is strong (+0.69 rising to +0.86) and the cue-frame read-out is intact. The invalid-cue cost has changed in kind, not merely in degree, becoming an effective blindness to uncued changes.

4.1 A biased-competition / normalization mechanism

This set-size scaling is what biased-competition and normalization accounts predict (Desimone et al., 1995; Reynolds et al., 1999, 2000; Reynolds and Heeger, 2009). With more competitors in the suppressive field, the top-down bias toward the cued location is harder to overturn: reallocating attention away from a strongly favoured cued item toward an uncued event requires overcoming a deeper pool of competition, and at set size nine that reallocation fails outright. The same logic frames the effect as a load-graded extension of the classical Posner invalid-cost (Posner et al., 1980, 1994), where the cost of a misdirected cue grows with the number of competing locations (Carrasco, 2011). A priority-map reading is consistent: the model commits to the cued location, and a more strongly committed map—more competitors, stronger normalization—is correspondingly harder to override, echoing collicular and priority accounts of selective commitment (Krauzlis et al., 2018, 2013; Zénon and Krauzlis, 2012).

4.2 A top-down versus bottom-up dissociation

The collapse is specific to the cue-gated top-down read-out. An element-wise-affine variant of the same architecture, which reads change bottom-up, does not collapse at set size nine: it continues to lock on the uncued change (invalid-cue excess $\approx +0.61$ at f5 and $+0.59$ at f6, comparable to its valid-cue excess of $+0.71 / +0.66$), because any sufficiently large orientation change pulls its attention by salience regardless of the cue. The price of that immunity is a weaker cueing benefit: a reader that is captured by salience cannot express the gain that a cue-gated reader derives from prioritizing the cued location. The dissociation localizes the catastrophic invalid-cue scaling to the top-down, cue-gated channel, and identifies it as the same channel that confers the cueing benefit in the first place—selection and its cost are two faces of one mechanism (Sprague et al., 2018; Herman et al., 2020).

4.3 A falsifiable behavioural prediction

The abolished top-down reallocation at set size nine predicts a large behavioural invalid-cue cost there: uncued changes should be largely missed rather than merely detected at higher magnitude. We state this as a prediction. More generally, the model forecasts that the behavioural invalid-cue cost in cued change detection should grow with set size—recoverable at low load, severe at high load—a directly testable claim for covert-attention experiments in primates and humans (Posner et al., 1980; Luo and Maunsell, 2019).

4.4 Limitations

Several caveats bound these conclusions. The set-size-nine attention read-out is a snapshot of a model under continued training, taken earlier than the set-size-four read-out, so the magnitudes there may shift as training proceeds. The set-size-nine behavioural psychometric has not been measured; the large invalid-cue cost at high load is therefore a prediction to be confirmed at convergence, not a measured result, and no behavioural number is reported for that condition. Change-lock is reported throughout as excess over the relevant uniform baseline (0.25 at set size four, 0.111 at set size nine) to keep the quantity comparable across set sizes. Finally, the findings rest on single trained instances of each variant; replication across seeds is needed to establish that the top-down-versus-bottom-up dissociation is robust rather than idiosyncratic.

5 Methods

Model. The agent is a convolutional JEPA trained end-to-end from raw pixels. A squeeze-and-excitation residual convolutional front-end (no variational autoencoder) produces per-location

features; a V-JEPA prototype-softmax self-distillation auxiliary objective provides a representational learning signal; a cross-attention memory reader (the cue-gated, top-down “crossattn1” module) attends from query tokens over spatial memory keys; a spatial xLSTM with memory width $d_{\text{mem}} = 128$ maintains recurrent state; and a distributional actor-critic head produces a per-frame `wait/change` decision. For the bottom-up control we use an element-wise-affine variant of the same architecture in which the memory reader is self-attention rather than cue-gated cross-attention.

Task. The agent performs spatially-cued, value-directed orientation-change detection. A coloured cue marks one location and signals a reward magnitude by colour (cue colour-values $[1, 3, 5]$); a validity proportion sets the cue’s spatial reliability. Each episode has seven logical frames; a change occurs on half of trials, always at frame 5, and is therefore detectable from frame 5 (f5) onward, with f6 one frame later. A trial is *valid* if the change occurs at the cued location and *invalid* if it occurs at an uncued location.

Set sizes. SS4 (task vda4): four stimuli on a 2×2 grid, 25×25 -pixel patches in a 50×50 image. SS9 (task vda9): nine stimuli on a 3×3 grid in a 75×75 image, with the cue at the top-left (P0) or bottom-right (P8) location. The architecture, auxiliary objective, and decision head are identical across set sizes; only the set size changes.

Change-lock metric. For each frame we read the cross-attention distribution over the spatial memory keys (averaged over queries) and compute the attention mass on the key corresponding to the changed location. We report *excess change-lock*: this mass minus the uniform baseline, which is $1/4 = 0.25$ at SS4 and $1/9 = 0.111$ at SS9. Reporting excess-over-uniform renders the two set sizes comparable. Values are reported at f5 (the change onset) and f6 (one frame later), separately for valid and invalid trials and for each cue placement (S1/S4 at SS4; P0/P8 at SS9). For the affine control the same metric is computed over the self-attention keys with the SS9 uniform baseline of 0.111.

Psychometric protocol. The SS4 valid-versus-invalid psychometric is measured by sweeping the orientation-change magnitude over $[0, 4, 8, 16, 28, 44, 64]$ degrees and recording $P(\text{detect})$ separately for valid and invalid trials. SS4 behavioural summary statistics (correct, hit, correct-rejection, false-alarm rates, reaction time, and the breakdown by cue colour-value) are computed over 600 greedy episodes.

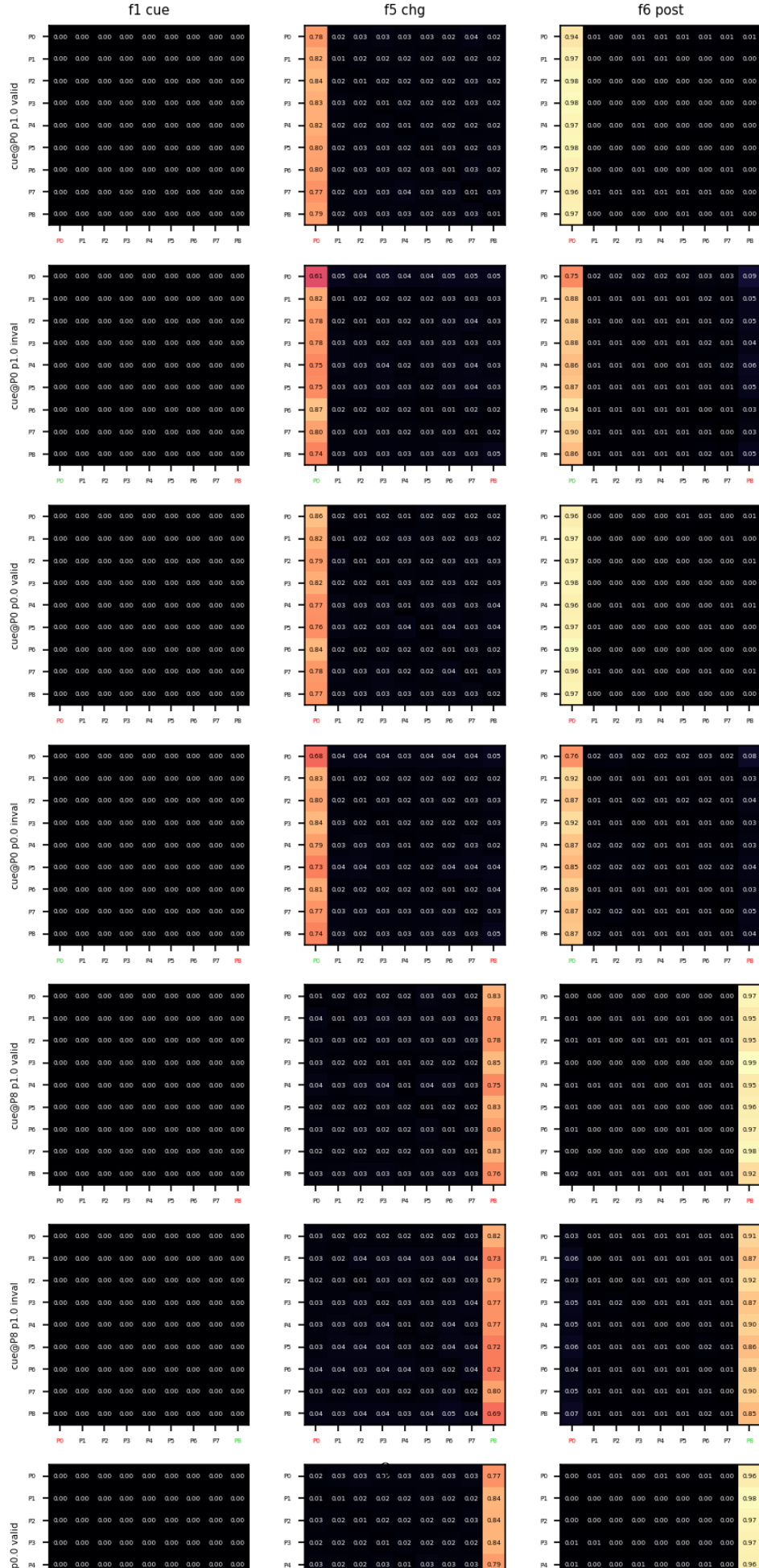
Snapshots and scope. Analyses use model snapshots at the following training iterations: SS4 at iteration 9599; SS9 cross-attention at iteration ≈ 4600 ; SS9 affine control at a comparable iteration (≈ 4400). The SS9 cross-attention model is a snapshot of a run still under continued training. Accordingly, the SS9 attentional results are reported as a mid-training snapshot, and the SS9 *behavioural* psychometric is **not yet measured**: the large behavioural invalid-cue cost implied by the abolished SS9 change-lock is stated throughout as a prediction to be confirmed at convergence, never as a measured result. All change-lock quantities are reported as excess over the corresponding uniform baseline.

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grid9 crossattn1 — MEMORY keys A[query→key] (mean/64); green=cued red=changed; iter 4599



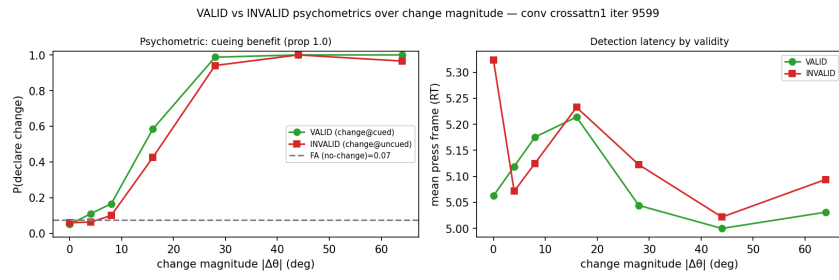


Figure 4. SS4 psychometric for valid versus invalid cues. The invalid threshold is shifted rightward (e.g. +0.16 at 16°) but both curves recover to ≈ 1.0 by 44° .

grid9 affine_ew — self-attn keys A[query→key] (mean/64); green=cued red=changed; iter 4399

